Research Plan
Significance
Migraine is a primary episodic neurological disorder that affects 1 in 5 people (660 million) in LMICs[1]. It is more prevalent than diabetes or chronic obstructive pulmonary disease[2,3] and its prevalence is rising with urbanization[4]. Most LMICs are undergoing rapid and chaotic growth in cities. Two-thirds of all of migraine patients and 73% of chronic migraineurs are women[5,6] who have a harder time in accessing healthcare in LMICs[7]. Migraine is acutely debilitating. It causes incapacitating headaches along with nausea, vomiting, fatigue, and extreme sensitivity to light and sound[8]. 90% of migraineurs cannot function and often need to retreat to a dark and quiet space for the duration of the attack[9-12], which can last between 4 and 72 hours[13], resulting in absenteeism from work, school, or social activities. Patients report the unpredictability of migraine attacks to be as distressing as the pain and disability associated with an attack[14]. The lack of control over keeping appointments causes anxiety that in turn augments the risk for a migraine attack, perpetuating a vicious cycle.

Migraine is difficult to manage in well-resourced societies. Medications that prevent attacks (prophylactics) and those that treat an attack (abortives) work to a certain extent for some of the migraineurs some of the time[3,15]. In LMICs, the inadequacy of partially effective treatments is compounded by the lack of access to medications and physicians of any kind, let alone neurologists or headache specialists[16]. For most patients in LMICs, identification and modification of triggers and helpers are the only means of managing migraine. Unfortunately, only 35% of migraineurs can predict an impending attack with >50% accuracy[17]. This approach requires patients to accurately identify factors as triggers or helpers, estimate their effect size and not assign them disproportionate weight if they occur just before an attack (recency bias). These fastidious requirements have led to significant discrepancies between results from retrospective diary studies and more formal investigations[13,18-25]. We seek to overcome these prior limitations by developing and testing objective approaches to personalized migraine trigger determination, risk assessment, and ultimately personalized migraine attack prediction. *These contributions are significant as they would empower patients to take more informed actions and acquire control over their condition, thereby reducing their migraine burden*.

Personalized risk assessment and prediction for individual migraine attacks hold significant promise for improving quality of life and reducing migraine-related costs around the globe. For most migraineurs in LMICs, tracking helpers and triggers is the *only* option to ease the burden of their disease. Predictions of sufficient accuracy and lead time would allow patients to make preventative lifestyle changes, thereby preventing an attack. If that is not possible, patients could clear their schedules and prepare for an impending migraine by taking medications early in the prodrome when abortive medications are most efficacious. Being able to predict the next migraine attack would also help allay the anxiety, fear, and sense of helplessness that comes with the unpredictability of migraine attacks. Accurate migraine risk assessment and attack prediction will undoubtedly change the current migraine prevention and management strategies and constitute an important advancement for this highly prevalent and incapacitating disease. Our work is aligned with the following priorities in the NINDS Strategic Plans 2021-2026[26]: Improving Treatments, Advancing Health Equity, and Inclusion and Diversity.

Innovation
Our previous work shows that patients' mobile diary data, digital biomarkers (behavioral or physiological data from wearables), and weather data can be leveraged to predict migraines with 86% accuracy, 24 hours ahead. Our prediction model worked well because we collected data longitudinally – across inter-migraine, premonitory and migraine periods – and built a personalized '*migraine map'* for each patient. To compare, single data stream models that utilize patient-reported stress as the only input can predict migraine attacks in individual patients with 65% accuracy[27]. Likewise, models that use phone diaries and periodic sampling of digital biomarkers and pool attacks from different patients achieve 62% predictive accuracy (AUC 0.62)[28]. We know from our own patient surveys that fewer than 25% of migraineurs would use a model with ≤ 65% predictive accuracy.

In our current proposal, we propose to augment the diary data with important and physiologically relevant digital biomarkers gathered via the patient's smartphone, since most migraineurs in LMICs do not have wearable devices. Digital biomarkers from smartphones are comparable to those from wearables in predicting stress[29]. The low/no literacy version of our digital diary allows us to reach migraineurs who cannot read or write. Our smartphone app uses the accelerometer, gyroscope, ambient light sensor, microphone, camera, and GPS chip in a patient's smartphone to unobtrusively gather longitudinal health-data with minimal effort on their part. Metadata from texts and calls and a digital diary app that asks the patient five or fewer simple questions augment the sensor data. Food is an important trigger[30-32] and helper[31-34] but has not been used in prediction because it is onerous for patients to document, and effective analysis is plagued with intra- and inter-personal variations, uncertainties about the timing and magnitude of effect, and correlation with other triggers and helpers. We have identified a subset of champion patients who are willing to photograph everything they eat with their phone camera and create the first ever real-time food library for a cohort of migraineurs.

The research proposed in this application represents a new and substantive departure from the status quo. Using a smartphone for unobtrusive, longitudinal, real-time, and objective data collection of environmental, physiological, and behavioral variables will allow us to develop the richest, most diverse migraine library built to date. The model will learn continuously and adapt to the changing course of disease seen in most patients over a lifetime. By deploying our scalable approach to hundreds or thousands of patients, we will leverage the benefits of network externality to identify migraine patterns across populations and use these patterns to accelerate and improve personalized migraine prediction. A far greater understanding of the heterogeneity within migraine populations will emerge from the model than is possible by any of the current approaches.

This approach, if successful, will fulfill mHealth's long-held promise of smartphones being a therapeutic tool for mobile users. Our approach will enable us to achieve four unique results: 1) We hope to capture new, previously undiscovered patterns in migraine patients as our model will be able to identify new correlations across different triggers and helpers and migraine events 2) We will calculate the effect size of each trigger/helper for each patient to increase awareness amongst the migraineurs about the magnitude of impact that potential triggers and helpers have on their migraine attack frequency, allowing the patients to take preventative measures 3) We will do feature selection to retain only the most significant triggers and helpers in our model, allowing efficient model training as we drop redundant features and 4) Since migraine is linked to urbanization, we will look at factors tied to cities, including commute times, population density, air quality, and light and noise exposure. This will allow us to recommend community actions to reduce the burden of migraine.

This innovation has impact far beyond the prevention and management of migraine and extends to other chronic diseases with episodic exacerbations (CDEE). The work will demonstrate the value of multi-modal data aggregation, including data from smartphones[29], in the real-time management of a disease with a clear and definite pain profile. By closing the loop through patient-facing interfaces and the development of personalized migraine maps, our work will demonstrate how patient behavior can be modified through personalized presentation of relevant data and algorithm predictions. Our preliminary data suggests that the infrastructure can be developed efficiently and that patients are eager to engage in actively managing their disease.

Past work

We have developed and deployed a scalable data-acquisition platform for longitudinal and unobtrusive collection of environmental, physiological, and behavioral data streams, as well as archiving patients' demographic and clinical information. The core of this data-acquisition platform currently consists of a smartphone application (*MigraineAid*) and a server infrastructure that allows for pooling and archiving of the acquired multimodal data.

Preliminary explorations

Most patients for our early studies came from Dr. Ali's internal medicine practice. She had developed detailed paper diaries for her patients with CDEE and reviewed these logs with the patient to identify the confluence of factors that led to a particular exacerbation and devise intervention strategies. This one-on-one pattern discovery required time, diligence, and patient cooperation as well as a healthy dose of luck to *a priori* identify the right set of triggers to track. While this approach empowers the patients to take informed actions affecting their health condition, it could not scale to large patient cohorts. Univariate regression analyses between trigger and migraine intensity yielded maximum correlation coefficients of 0.4, suggesting – not surprisingly – that no single trigger correlates strongly with migraine severity. We developed the digital diary in MigraineAid to overcome the drawbacks of paper diaries, and the diary data predicted migraine episodes with 63% accuracy 24 hours ahead.

MigraineAid

MigraineAid functions as a customizable electronic migraine diary that prompts patients for daily migraine updates and to log migraine-associated symptoms, triggers, and helpers (Figure 1). Our patients helped us develop the app by confirming the need for and identifying the most desirable features in a digital diary. They reviewed the currently available electronic migraine diaries for us, and a series of focus group meetings provided iterative feedback on successive candidate designs, layouts, and functionalities. We wanted a design that would increase a migraineur's acceptance of and willingness to interact with the app, particularly in the run-up to, during, and immediately after a migraine attack. Our focus group unequivocally favored peaceful, darker colors and simple page layouts to reduce photo-stimulation and symptom exacerbation during a migraine attack.

In its initial encounter with the patient, the migraine app guides the patient through a series of screens, and gathers demographic information, comorbidities, current medications as well as the nature and location of headache. It also asks for typical headache duration, migraine-associated symptoms, triggers, and helpers. For each piece of information, patients can pick options from a predefined list or provide their personal choice through data-entry fields. Additionally, the patients can choose the number and time of their daily prompts. These prompts appear as text-message reminders that link to the update screens (Figure 1 b-e) that ask the user about their day (Figure 1b), any migraine attacks for that day (Figure 1c), triggers (Figure 1d) and helpers (not shown).

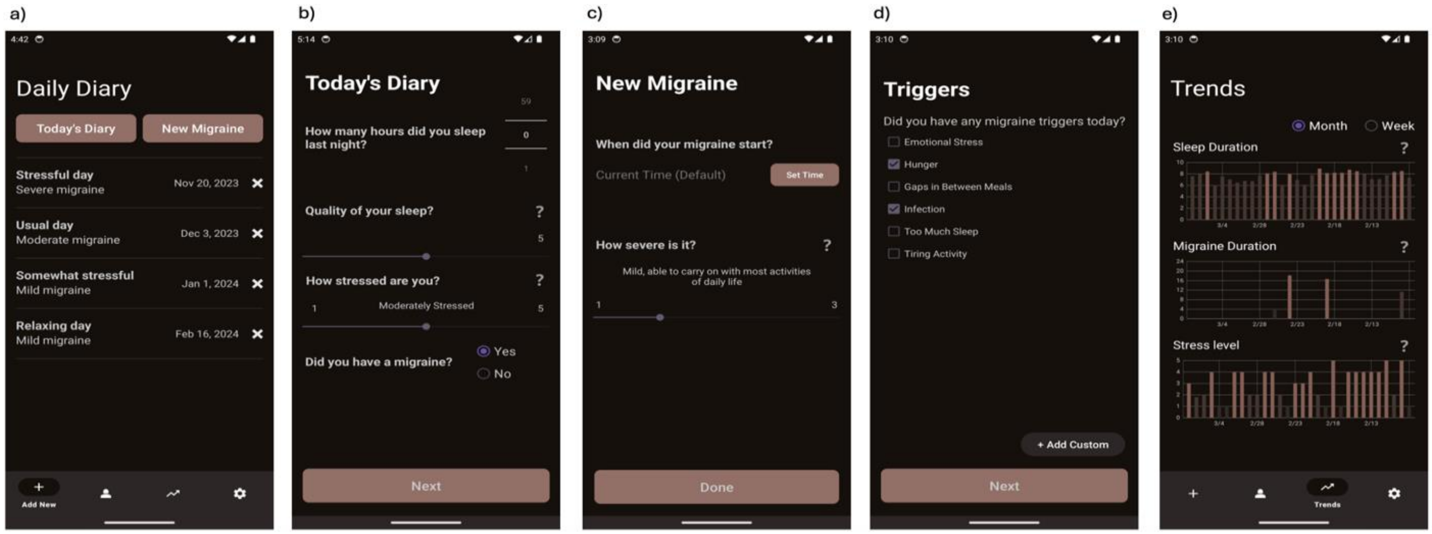


Figure 1: MigraineAid daily update screens. Home screen (a), Patients are prompted to provide daily updates on their sleep, stress, and migraine status (b), new migraine (c), and triggers (d) and are presented with the trend of their entries (e). Note that the color scheme was chosen after extensive focus group discussions with migraine patients to reduce stark contrasts and thereby photo-stimulation during a migraine attack.

The app provides feedback to the patients through a trend screen that summarizes their sleep, stress, and migraine patterns over the past week (Figure 1e). Upon completion of a daily update, the app data is uploaded to a Google Firebase server. We have instituted state of the art security protocols for data in the Firebase server.

Environmental data

MigraineAid connects to the phone's location tracker to log and upload to the Firebase server time-stamped location information every 15 minutes or more frequently when significant location changes are detected. With this information, weather data can be retrieved by connecting to and querying the National Oceanic and Atmospheric Administration's (NOAA) weather archive. Available weather information includes parameters that have been implicated as migraine triggers, such as barometric pressure, temperature, or relative humidity[35-37].

Physiological and behavioral data

We used Empatica E4 Wristband[38] to track physiological (heart rate, electrodermal activity (EDA), temperature, pulse plethysmogram) and behavioral (steps taken, sleep) data streams (Figure 2). We have developed the infrastructure to merge data collected from the E4 with our MigraineAid data at our server backend.

Server architecture

MigraineAid integrates the app-based data and location information and sends the resultant data to a Google Firebase server. Empatica E4 uploads data to its own servers and we downloaded it daily and merged the data streams with the MigraineAid data on our own server. We retrieve weather information from NOAA database via the Weatherbit application program interface (API).

Pilot study in the US

Five champion patients wore the Empatica E4 wearable device for 3-10 months for the pilot study of our data collection and archiving infrastructure. The E4 tracks temperature, EDA, pulse plethysmogram, and heart rate.

It also contains a three-axis accelerometer for movement tracking. Patients provided daily migraine updates via the MigraineAid phone app (Figures 1 b-e), including hours and quality of sleep, stress score, migraine events, triggers, and helpers as well as location information. This dataset covers 1071 patient-days with 733 participant entries over 656 days and includes156 lurking migraines and 204 migraine attacks. The resulting library predicted migraines for 4 out of 5 patients with 86% accuracy 24 hours ahead. One participant had no migraine episodes during the study. Figures 2 and 3 show 24 hours of data from one of our patients.

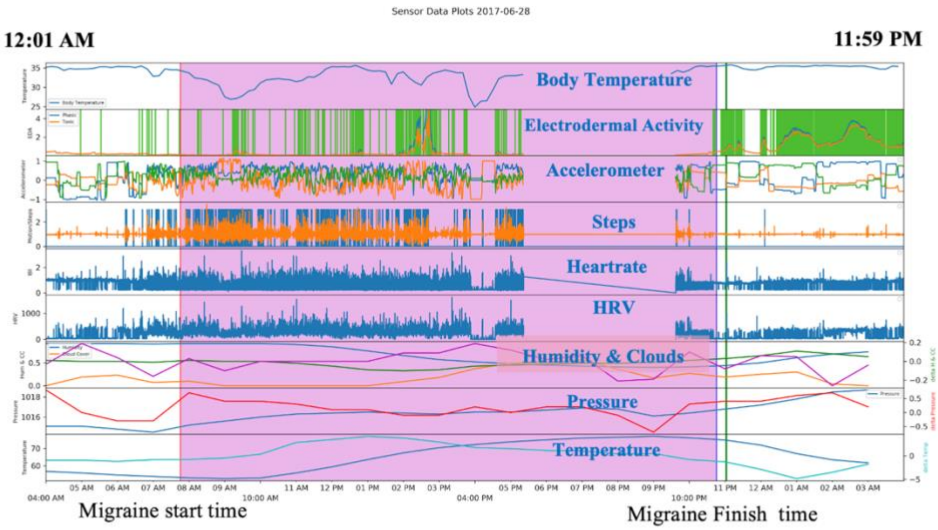
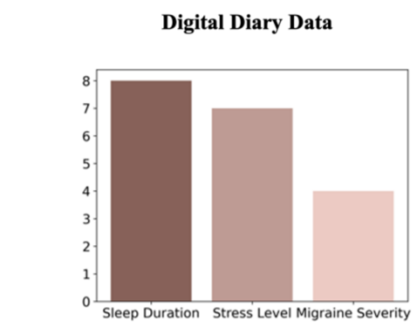


Figure 2: We collected data on the body temperature, electrodermal activity, and heart rate from Empatica E4. In addition, we collected phone accelerometer data to calculate the steps and used the GPS to extract weather data. The region in pink shows the duration of a migraine attack in a 24-hour window, with this episode lasting from 8 am till 11 pm.



Triggers Today	Symptoms Today	Helpers Today
Emotional Stress	Fatigue	Massage

Figure 3: On the same day as in Figure 2, this user slept for 7.5 hours, had a stress level of 7 on a scale of 10, and a migraine severity of 4 on a scale of 5. The user has mentioned migraine triggers, symptoms, and helpers.

Pilot Study Roll out in Pakistan:
Learning from our work in the US, we started recruiting English speaking patients in Pakistan in 2023. Very few Pakistanis use wearable devices, but Pakistan has 46 million migraineurs[39] and 128 million mobile-broadband subscribers[40], 94% of whom use Android devices[41]. Digital biomarkers from smartphones are comparable to those from wearables in predicting stress[29]. We are using the smartphone as a data collection and modeling hub. Android APIs allow us to collect phone sensor data from the accelerometer, microphone, and ambient light sensor. We have data coming in from 28 patients right now.

APPROACH: R21 PHASE

Approach – *Specific Aim 1: To update and optimize the existing infrastructure of our migraine library for data collection in LMICs.*

Introduction – Over the last year, we have built an android version of MigraineAid by incorporating feedback from our users in Pakistan and are currently working on a low/no literacy version of our digital diary. By expanding on our prior work, *the central goal of Aim 1 is to adapt our data collection platform for LMICs and to develop associated data validation modules to increase the coverage and quality of the data in our library.* To improve data coverage in areas with poor internet connectivity, we will enable on-device storage in the event of a wi-fi interruption. We will develop an interface (migraine map) that meaningfully and simply conveys the insights gained from the migraine prediction model (Aim 3) to the patient via the phone screen. We will optimize our data architecture for rapid scaling and develop automated quality assurance (QA) modules that monitor data coverage and its quality in real-time and generate notifications of potential data integrity problems to study staff and participating patients.

Research Design

A. *Developing a low/no literacy app*

   One in two Pakistanis (115 million people) cannot read or write[42] and will never meet a physician *ever* in their lives. We are developing a version of MigraineAid without text to make it accessible to them. We will substitute text with intuitive pictograms and use audio input and output for communicating with the patients in their native language, Urdu. The audio input and output will use Phonetically Rich Urdu Speech Corpus (PRUS) as its foundation[43].

Dr. Raza (PI) is the principal developer behind PRUS. We will finetune PRUS to produce more natural voice outputs. The app will communicate the risk of an impending attack to the patient with the help of a dynamic graphic map. The map will use colors and shapes rather than numbers and text to convey the risk of an impending attack to the patient (Figure 4).

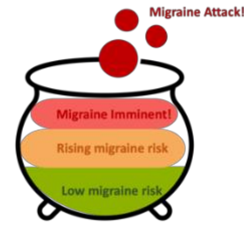


Figure 4: Color and shape-based migraine map

Expected outcomes – By developing a low/no literacy app, we will reach a larger and more diverse patient base and offer help to those patients who have help from no other source to manage their disease.

Potential problems & alternative strategies – The intended patient user base for the low/no literacy app is not monolithic, and developing an app that works for most of them will require multiple iterations with feedback from patients who are true representatives of this group. We will hold iterative focus groups and utilize A/B testing to identify the optimal user interfaces for the app.

B. *Quality assurance (QA) module development*
The QA modules will consist of a set of data integrity and signal-quality checks that act on the incoming data streams to assess and quantify the fidelity and frequency of the incoming data streams. These modules will produce daily per-patient summary statistics so study staff can dynamically monitor data completeness and patient retention. Additionally, these modules will restrict user input to valid ranges, preventing nonsensical data. If a patient's incoming data traffic seizes or drops significantly below our data completeness target, the User Notification and Feedback module will send an automated notification to alert the patient and study staff. The Notification module can also be used to provide positive feedback to the patient for reaching individualized personal target goals (such as maintaining adequate quality sleep or keeping up with one's migraine helpers), thus providing positive reinforcement to the patient.

We will enable on-device storage of data in the event of an internet interruption. When the app reconnects to the internet, we will use background synchronization to automatically update data and ensure consistency with the server. Once the server is updated, the on-device data copy will be deleted. This allows users to switch between online and offline modes without manual intervention.

Expected outcomes – The expected outcome of this task is increased data integrity and signal quality with automated data validation. These efforts will enhance patient engagement and retention through real-time feed-back and personalized goals.

Potential problems & alternative strategies – The development of the QA, User Notification and Feedback Modules is straightforward. There will be trial-and-error in understanding the frequency of notifications sent as reminders for the users to add the diary data. We will be doing focus groups every 4 months to keep up to date on our user preferences, so that we do not send them too infrequent or excessive notifications.

C. *Revised hardware modules*
To achieve the ambitious goals set out in this proposal, we require a flexible database and compute architecture that supports a variety of real-time applications on and manipulations of the dynamically growing migraine database. Examples include dynamic accommodation of incoming data traffic from increasingly larger patient pools and deployment of modules and APIs for data-quality assurance, automated user notification, machine learning (ML) based prediction, and database query and manipulation (Figure 5).

To expand upon our existing infrastructure, we will exploit our eight-node, on-campus compute cluster (40 TB storage capacity, out of which 10 TB is currently available, 128 GB memory per node, 192 high-performance processor cores, out of which 72 cores are available). As demand increases, we will move to Cloud GPU and CPU. The pipeline on CPU will spin up an individual instance for each request. The pipeline on a single GPU will batch together data from 64 users in parallel and perform inference in real-time. Our platform will contain one central application stack consisting of a Firebase (NoSQL) database layer for data archival and aggregation because of its flexibility, particularly when new patients are added, and another for application execution (data wrangling and analysis). An API server and event router will provide the middle-layer between the cluster and communication modules, the NoSQL application layer, a website frontend, and a set of APIs to expose the storage and compute server functionality to patients and collaborating researchers.

This application stack will 1) serve as an interactable repository for all collected measurements, diary entries, annotations, and analysis products and results; 2) serve as a computationally scalable processing platform to perform in-place analyses directly on collected data; 3) present collected personal data, anonymized user-base-wide data, and analysis results in a comprehensive, user-friendly format in the mobile application; 4) allow users to easily add and annotate their personal data sets through the patient-facing mobile app interfaces; and 5) promote data openness by allowing users total access to their personal data set in open formats, such as CSV for tabular information and plain text for textual or narrative information. Our group has familiarity with the implementation of each one of these stack elements.

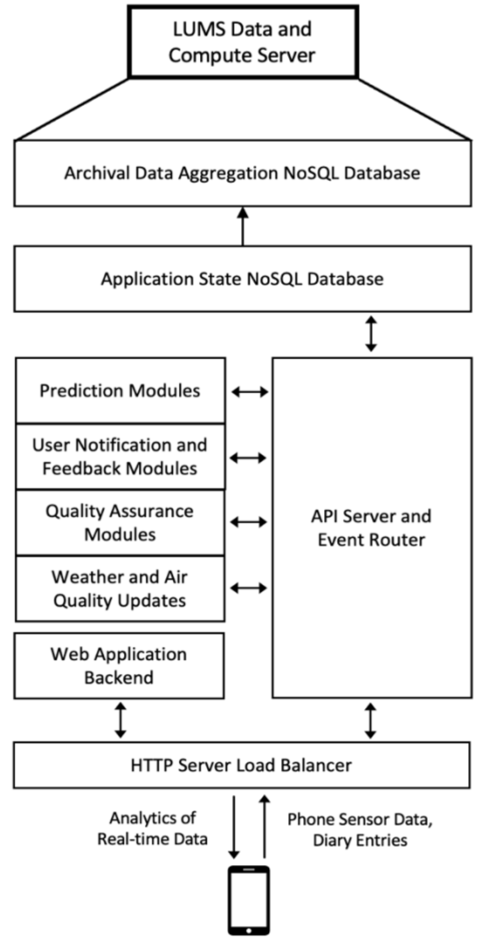


Figure 5: Data and compute server architecture as a central application stack.

To ensure data privacy, we will implement stringent protocols, including encryption at rest and in transit, role-based access control, and regular security audits. We will anonymize personal identifiable information before storage, and strictly comply with HIPAA and GDPR regulations throughout data collection, storage, and analytical processes. We will get informed consent from each patient, granting them control over the use of their data, and ensure transparency in data handling.

Expected outcomes – The expected outcome of this task is a scalable health library platform and associated server infrastructure to simultaneously receive, process, and archive multimodal physiological, environmental, and clinical information from thousands of patients and to provide patients and collaborators with API-based access to the database for query, download, and annotation.

Potential problems & alternative strategies – Given the large design and parameter space of the application stack, a key challenge in the development of the database infrastructure is defining which refinements, augmentations, and extensions will be in scope over the five-year project horizon. We will opt for robust implementations that are stable under extensive load testing and prioritize design options that enhance the user experience for increased patient engagement and retention.

Approach – *Specific Aim 2: To build a health library that collects, organizes, and audits longitudinal data streams of personal and environmental factors for each migraine patient in real time by using the sensors and devices in a mobile phone as data collectors.*
Introduction – We have used wearables like Empatica[38] and Everion[44] to collect physiological data from migraine patients in the past. Very few patients in LMICs use wearables and thus *the central goal of aim 2 is to use the sensors and devices in a smartphone as the main source of patient data.* Digital biomarkers from smartphones are comparable to those from wearables in predicting stress[29]. We will use the phone accelerometer for measuring the patient's physical activity and commute. The phone GPS chip will help us acquire weather and pollution data. We will also use android phone metadata (number and timing of calls and texts, screen time, app interaction) to gauge each patient's stress[45]. We will also collect new data streams by utilizing the phone ambient light sensor and microphone for estimating the patient's sensory load[46]. A subset of champion patients will take photos of everything they eat and upload the photos to build a food library for migraine prediction[29-33].

Research Design:

A. *Use phone sensors (instead of wearable devices) for collecting physiological data (Figure 6).*
All smartphones have accelerometers which we will use to track physical activity. Commuting is a significant risk factor for migraine[47], and we will use a combination of GPS and accelerometer data to find the time and

distance of patients' commutes. We will also use the GPS chip to pull weather and pollution data from NOAA and IQAir.com respectively. Air pollutants like Particulate Matter and nitrogen dioxide have been linked to migraine in multiple countries[48]. We have used EDA and Heart rate sensors in wearable devices to gauge stress for individual migraine patients. We will now use android phone metadata (number and timing of calls and texts, screen time, app interaction) to measure patient stress states and use both the raw data and stress estimates as input for migraine prediction[45]. We will access third party messaging apps like WhatsApp data with the patient's permission so that the metadata we have is a true reflection of the migraineur's stress state[49].

Expected outcomes – The expected outcome of this task is a large database of digital biomarkers collected via patient smartphones rather than wearable devices. Since Pakistan has 128M smartphone users, this aim will greatly increase the pool of migraineurs we can collect data from and allow us to build a large and rich dataset for predicting and preventing migraines.

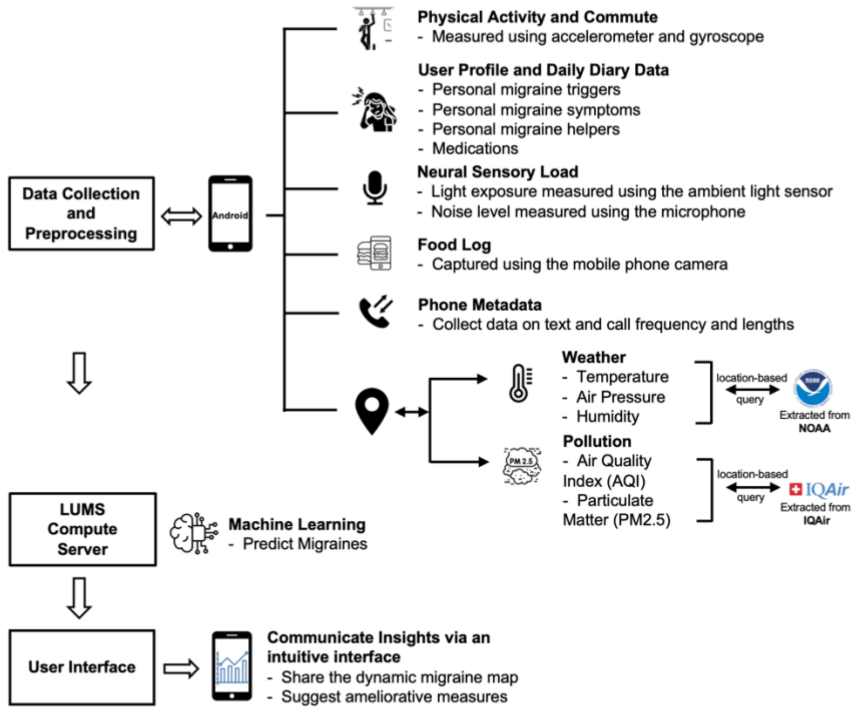


Figure 6: Revised data-collection and server infrastructure.

Potential problems and alternative strategies – Android phones vary a great deal in the fidelity of their sensors. We have encountered this problem while using wearable devices in our previous work and will group incoming data by phone type and learn early in the study which smartphones have more reliable sensors and adjust the labeled input for ML models accordingly.

B. *To collect novel data streams by utilizing smartphone sensors and devices*

To augment the predictive and preventative capacity of our model we will add novel data streams to our database. To estimate the sensory load experienced by a migraineur, we will periodically sample the ambient light and noise she is exposed to with the ambient light sensor and microphone in her smartphone. To measure the impact of the patient's diet on her migraine burden, we will ask a subset of champion patients to take photos of everything they eat and upload them to our server. For the first year, the patients will verbally describe what they are eating so we have a labeled food library. We will use unlabeled food photos for prediction in quasi real time. Once the labeled food library has matured, we will add its variables to explore its impact on migraine prediction and prevention.

Expected outcomes – The expected outcomes for this task are rich datasets that will delineate the impact of sensory load and dietary choices on migraine burden.

Potential problems and alternative strategies – Asking patients to take photos of everything they eat is onerous for the patients. Only patients with high migraine burden will be motivated to join the subset collecting this data. We will focus on identifying the patient subset that benefits from tracking their diet as early as we can and only ask those patients who benefit from this task to invest effort into it.

Approach – *Specific Aim 3: To generate real-time personalized migraine predictions for each patient and in time, identify factors that might be common helpers or triggers for certain subgroups of migraineurs.*

Introduction – Our pilot study has shown that we can predict migraine episodes with 86% accuracy 24 hours ahead by taking the harmonic mean of the predictions made by the diary and the wearables data. Weather data did not improve prediction for the patients in our pilot study. Our goal in aim 3 is to increase the lead time to prediction and improve its accuracy while replacing the wearable data streams with those coming from the smartphone. Our own survey of 106 migraineurs shows that patients want 24-48 hours of notice with 85%

accuracy and prefer a model that produces false warnings rather than miss an episode. *The central goal of Aim 3 is, therefore, to identify personalized migraine triggers, estimate their effect size, and predict attacks with a lead time of at least 24 hours with ≥ 85% accuracy. Once the data library is large enough, we will pool the anonymized data to identify factors that may be common helpers or triggers for certain subgroups of migraineurs.*

Research Design:

A. *Personalized trigger/helper determination and migraine prediction*

We have used Recurrent Neural Networks (RNN), Hidden Markov Models (HMM), and Long-Short-Term Memory (LSTM) with decent predictive results so far, and in the R21 phase of this study, we will continue to use these models. The main limitation of these models is that they are highly impacted by the recency of helpers and triggers and may miss factors that happened a week or fortnight before the attack.

Migraine triggers and helpers affect the physiology of the migraineur for varying lengths of time. For example, air pollution impacts the patient in perpetuity, the menstrual cycle for weeks, a thunderstorm for a couple of days, and noise and bright lights for varying periods of time. The appropriate ML model for data streams with variable impact-time lengths is a Transformer model[50]. Transformer models can accommodate factors that had an impact on the migraine cycle hours, weeks, or months ago. However, Transformer models require large amounts of training data, and since each patient's predictions come from their individual data library, we are not going to use a Transformer model right away. A migraine attack arises from an accrual of helpers and triggers over a 'receptive' timeframe which varies from patient to patient[51]. For example, if the receptive timeframe is one week for a group of patients, a Transformer model may need data for 50 weeks from 100 such patients (5000 patient weeks) before it can offer meaningful predictions. We will recruit 500 patients and will have 10,000 patient-weeks of data by the end of year one and will start using the Transformer for most of our patients in year 2. We will be using Transformer models exclusively in R33 phase of this work.

Further, we have successfully used random forests and transformers in our work to identify minimal subsets of features with the highest predictive power. In the case of transformers, this is achieved using the self-attention weights to identify the most attended features in a trained transformer model[52-53]. Dr Sheikh (co-investigator) will review these top-k highest predictive features to sift true triggers from red flags.

In our previous work, we have used a probability of greater than 50% for a particular event as the threshold where it is declared as an impending migraine. In addition to providing patients with the (binary) migraine attack prediction, we will also give them the underlying probability and how the various triggers contribute to the overall probability. This output will be displayed to the patient as an intuitive, graphic migraine risk map (Aim 1) and updated regularly as new information becomes available (e.g., changing weather forecasts).

A key advantage of our data-collection approach is that we collect data during the premonitory, migraine, and inter-migraine phases of the migraine cycle. This allows us to use a patient's own inter-migraine period as a control for the trigger/helper determination.

B. *Identify factors that might be common triggers/helpers for certain subgroups of migraineurs.*

Once we have a large pool of data from thousands of patients, we will analyze the anonymized data to see which factors are common to migraineurs in a certain geography, age range, or gender. If the effect size of certain triggers and helpers can be determined for certain subgroups of patients, we can reduce the length of time the model would need data from new patients in that group before it can make accurate predictions.

Validation approach – We will initially follow a standard approach to train and test the prediction model on separate segments of a patient's data. To minimize patient anxiety, we will initially run the model in 'silent mode.' In silent mode, ML algorithms act on data in real-time, but the results are recorded rather than displayed to the patient. Once the predictions have ≥ 85% accuracy, the results will be shown to the patient.

Figure of merit - We seek to achieve migraine prediction with true positive and true negative rates of over 85%, as both falsely predicted migraines (false positives) and falsely missed migraines (false negatives) have the potential to induce anxiety and may contribute to a patient's migraine burden.

Expected outcomes – The expected outcome of this task is an objectively determined, patient-specific trigger/helper profile for each migraine patient and later a stratification of certain triggers and helpers into common factors that affect certain subgroups of migraineurs.

Anticipated problems & alternative strategies – We will need a sufficiently large number of migraine events per patient to be able to meaningfully use the ML models outlined above. While we will develop the largest and most diverse migraine library, how many attacks we will be able to record depends on patient recruitment and retention (hence our focus on patient engagement in Aim 1 and 4). We will do our best to recruit the same patients to both waves of validation, which would allow us to collect 60 months of data in some patients.

APPROACH: R33 PHASE

If specific milestones are met, the R33 phase will optimize the model for wide distribution and ease stakeholders' access to the library.

Approach – *Specific Aim 4:* To optimize the interface of the model for widespread distribution and develop a diverse cohort of users in multiple neighborhoods.

Introduction – Our previous work in the US and Pakistan has made it clear that the development of an effective migraine database and prediction tool is only possible through an ongoing commitment to patient engagement. Recruitment of successively larger cohorts of patients to provide data to the library will make the database more valuable for migraine prediction and knowledge discovery. Equally important is the feedback from these cohorts on which library functions will enhance their engagement and help them manage their disease better. *The central goals of Aim 4 are therefore to engage successively larger cohorts of patients to populate the migraine library and to provide critical feedback on how the value of the library can be further improved.* To address this aim, we will roll out yearly validation studies and solicit detailed patient feedback through one-on-one phone conversations with the project coordinator, focus groups, and web-based surveys. Our experience has taught us that different feedback channels yield overlapping but distinct information that is critical for developing the most engaging interface and for maximizing the migraine app's value to the patients. The value of the migraine library grows in direct proportion to the number, diversity, fidelity, and completeness of the data collected.

Research Design

A. *Wave I Data acquisition and study design* – In years 1-2, we will recruit 500 patients in each year to participate in Wave I of our data collection and library validation study. Our patients will provide digital diary updates via the MigraineAid app as well as behavioral data streams via smartphone sensors. Patients will be eligible for enrollment into the study if they have a diagnosis of migraine according to the International Classification of Headaches criteria, are older than 18 years, possess and regularly use a smartphone. Patients will be excluded from the study if they have conditions that preclude them from interacting with a smartphone at least six days/week. The first wave of patients will be drawn from Sughra Shafi Medical Complex (SSMC) and Shalamar and Indus hospitals (letters of support attached) and educational institutions around the country. We will hold seminars in universities to facilitate recruitment for the study. The project coordinator will reach out to every study participant at 3-month intervals, and we will lead focus groups every 4 months. As part of Wave I, we will test the automated review of data fidelity and integrity and the automated feedback loops in the library's architecture and gain patient feedback on library interface design and ease of use.

   Number of migraine events and data completeness – Since an average migraineur has 2.8 attacks per month[3], we expect to capture around 16800 migraine events in year 1 and 33600 in year 2 during this wave of data collection. Based on prior experience with dataset collection, we define data completeness as 25 days of submitted migraine diary information for every 30 calendar days (83% coverage) and 144 hours of smartphone data every week (86% coverage). Our goal for this wave is to attain 90% or more data completeness for each patient (75% of all possible data for diary entries and 77% for smartphone data). Thirty-day user retention of mobile health app users tends to be well over 80%, especially when the health app is recommended by a physician and helps with a definite pain point[54], and hence our patient retention goal in this first study wave is 80%. As a buffer for dropouts, we will have a pool of substitute patients to recruit if a drop-out happens to occur early in Wave I.

   Expected outcomes – As stated above, our expected outcome is 80% patient retention with at least 90% of aggregate data completeness per patient over the duration of the two-year study period.

Anticipated challenges & alternative strategies – Timely feedback to the study participants if a data-stream is interrupted due to technical or personal reasons is likely the biggest challenge in this phase. We are planning to address this challenge by automating data integrity and signal-quality checks. To optimize retention, we will conduct monthly check-in phone calls and host focus groups to foster a sense of community. We will also maintain a reserve patient pool as a buffer against dropouts.

B. *Wave II*

Data-acquisition and study design – In years 3-5, we seek to recruit 1000, 3000 and 5000 patients in successive years to end up with a cohort of 10,000 patients at the end of year 5. The patients will provide daily migraine updates, and smartphone sensor data. We will engage a marketing partner with experience in recruiting for healthcare to help us recruit patients for this wave. A study team member will connect with every study participant once a year to solicit feedback on the patient's participation. We will hold focus groups every 4 months. There will also be a patient mixer every 6 months to build a sense of community and give face-to-face feedback to the research team. The mixers will be the catalysts for online patient communities that will drive recruitment and retention. We will make a short documentary in local language(s) to distribute via social media. Most of the patients in years 3-5 will come to the app via word of mouth and social media.

Number of migraine events and data completeness – We expect to record about 67200,168000 and 336000 migraine events in years 3, 4, and 5 respectively. Our data completeness goals are the same as in Wave I. Our retention goal for this wave is 65%. We will maintain a reserve patient pool as a buffer against dropouts.

Expected outcomes – Our expected outcome in this wave is 65% of patient retention with at least 90% of aggregate data completeness per patient over the duration of the study period. Additional expected outcomes include detailed patient feedback on their experience and how engagement with the health library, patient-facing user interfaces, and data integrity alerts can be improved.

Anticipated challenges & alternative strategies – Patient retention and maintaining a level of data completeness with minimal human prompting will be the main challenge of this phase. The main thrust of our work between Waves I and II will be the final optimization of the interface design to engender a bond with the patient. Interactions between member of the migraine community will be actively fostered via patient mixers, and we expect that these activities will increase patient retention and data uploads. Rigorously paring down the number of variables needed to reliably predict and prevent (Aim 3) and emergence of online communities of migraineurs based on geography will increase patient retention and data uploads, thereby increasing the accuracy of prediction and prevention, setting off a virtuous cycle of growth that will allow us to benefit from Network Externalities in wave II.

Approach – *Specific Aim 5: To wrap a Large Language Model (LLM) layer around the data library and build a Chat Interface that allows patients, physicians, and researchers to interact with the library in natural language.*

Introduction - A migraine library is as useful as the ease with which it can help migraineurs, their physicians, and the researchers working on the disease. *A primary goal of our work is to stimulate research into migraine risk assessment and attack prediction and to challenge the engineering and clinical communities to develop novel approaches to migraine management and prevention. To facilitate the full utility of our library by all stakeholders, we will use an LLM to build a chat interface that allows patients, physicians, and researchers to ask the migraine library complex questions in natural language*. The chat interface will provide real-time and customized responses, allowing stakeholders to leverage the data library for faster decision-making and research. Robust utilization of the migraine library will encourage all stakeholders to contribute more data to the library.

Research Design:

A. *Development of a customized LLM envelope for our data-library*

We will use an open-source LLM like Llama or Mistrel as a foundation and train it with migraine-related question-answer pairs that we will solicit from patients, physicians, and researchers. We will host the LLM locally to protect patient privacy. To build a rich pool of curated question-answer pairs, we will ask patients at focus groups and through MigraineAid, to send us questions they would like to ask an 'expert' about their disease. Drs. Sheikh and Ali will respond to these queries, and we will ask the patients to validate the expert replies. In years 3-4, we will ask a cohort of 5 physicians and 5 migraine researchers for lists of questions they might have for a migraine 'expert' and Drs. Sheikh and Ali will answer these questions. We will ask physicians and researchers to validate the expert responses. Since most LLMs have a chat function already built into them, we will need 1000-1200 question-answer pairs to train the foundation LLM for our purposes. These curated question-answer pairs, coupled with the data in our library, will allow all stakeholders to 'chat with our library' in normal human language and get insightful answers to their queries.

Expected Outcomes
The expected outcome of this task is a curated question-answer set of at least 1000 questions from patients, physicians, and researchers.

Anticipated Challenges and alternative strategies:
Gathering the questions is straightforward and we will offset the experts' biases by having the inquirers give feedback on the quality of the responses. Since LLMs can hallucinate, we will use Retrieval Augmented Generation (RAG), in which the LLM will be limited to using the migraine library as its source. We will also train the LLM to translate inferential queries into Python code so that its inferences are precise and cogent.

B. *Chat Interface Development*
We will build an intuitive interface to facilitate seamless interaction between all stakeholders and the LLM-powered library. The interface will be available both via the app for patients, and via a dashboard accessible by researchers and doctors. We will employ user-centered design principles so that all all users, regardless of their technical proficiency can use the migraine library. The design process would involve testing phases with real users to identify and rectify usability issues, including diverse focus groups to ensure the user interface meets their needs and expectations. Feedback mechanisms will be embedded within the interface, allowing users to report issues that will trigger audits in case the expected response to a query was not returned. This continuous feedback loop will enable iterative enhancements, ensuring the interface remains effective and user-friendly over time. We will protect patient privacy by restricting access to identified data to only patients and their treating physicians. We will credential all chat interface users and access to insights from anonymized data will be available to all stakeholders.

Our goal is to have at least 10,000 library users in 5 years. We are focused on Pakistan, but MigraineAid will be available on Google Play globally and we expect patients from a variety of regions to download the app and use it for managing their migraine. Using our anonymized data pool, we will build a foundation model that researchers in countries and regions other than Pakistan will be able to finetune with their local data and have it available for their patients for quick and accurate migraine prediction.

Expected Outcomes
The expected outcome of this task is an LLM-powered, intuitive chat-interface that will act like a competent research assistant for all stakeholders. This will increase stakeholder engagement by removing barriers for complex data queries. Active library use will increase motivation for active data entry which in turn will improve the accuracy and relevance of the answers the library produces, thus engendering a virtuous cycle.

Anticipated Challenges and alternative strategies:
Data security is a risk in opening the chat interface to all credentialed stakeholders, but we have interwoven state of the art security into the library from the ground up and have protocols for protecting the data and handling its breach if that were to happen.

Statistical Considerations:
<u>*For personalized prediction for a patient*</u>, sample size, 'n' is the number of migraine events that the Migraine Prediction Model (MPM) must analyze before it can identify the triggers and determine their relative importance for that patient. In our pilot study, we had an average of 51 events per patient (range 9-99) and by assigning 64%, 16%, and 20% of events to training, validation and test datasets (Pareto principle) we were able to predict attacks with 86% accuracy 24 hours ahead. The number of migraine events needed for useful prediction varied between 9-18 events in our data, but we had only 5 patients. An average migraineur has 2.8 migraines/month, our goal is to get at least 3-6 months of longitudinal data for each patient (Aim 4). Patients with seasonal triggers will need longitudinal data for longer periods.

<u>*To identify factors that might be common helpers or triggers for certain subgroups of migraineurs.*</u>
Migraineurs can be sub-grouped by genetics or phenotypes[55-58]. Subgroups of migraineurs may have similar 'sets' of triggers and helpers. This is a pilot study to explore if we can stratify migraineurs into subgroups by common sets of triggers/helpers and determine the components of these sets (Aim 3). Taro Yamane method calculates that for 46 million migraineurs in Pakistan, accepting an error rate of 1%, we need 1000 patients to be confident we will capture the subgrouping if it exists. We will have 10,000 users by year 5 and will use latent class segmentation and hierarchical modeling to identify migraineur sub-groups.

Milestones:
Milestone decisions will be made after discussion with the NIH program staff. Proposed R21 milestones are:

<u>*Milestone#1*</u>: Recruit 1000 users to the migraine prediction model (MPM) in conjunction with our clinical network.
- <u>*Rationale:*</u> Of the 100 users recruited for our model in Pakistan so far, ~30% remain faithful to the app for more than 3 months. We need 1000 users in R21 to retain sufficient users who will input > 3 months of data.

<u>*Milestone#2*</u>: Retain 300 users with 6 months of data with 80% completeness.
- <u>*Rationale:*</u> Personalized migraine prediction requires longitudinal data. To build and prove a robust MPM we need to retain migraineurs who are willing to give us ≥ 6 months of longitudinal data.

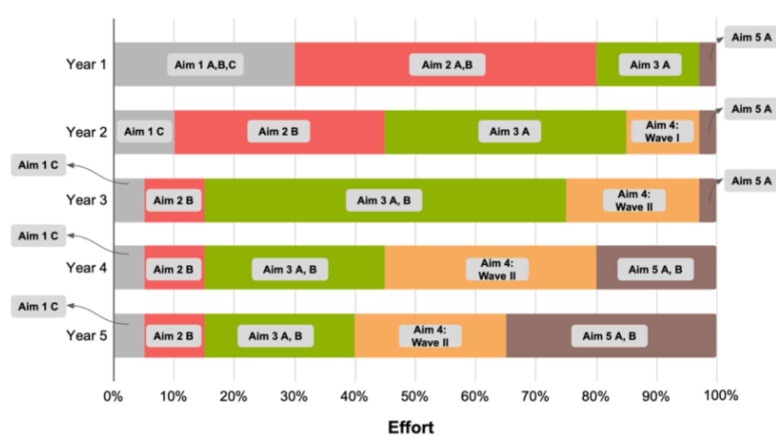
<u>*Milestone#3*</u>: Predict migraine with 85% accuracy for 33% of users with 6 months of data with 80% completeness.
- <u>*Rationale:*</u> Our own migraineur survey tell us that users want at least 85% accuracy from an MPM. 8% of migraineurs have chronic migraine (≥ 15 migraines /month) and the model is unlikely to work for them[59]. 20-30% of our patients will have complicated migraines or co-morbid conditions which will require medical care before the model can be useful for them[60]. The model has the potential to help 62-72% of its users and if it benefits half of this potential group in R21, the study will continue into the R33 phase.

Future work
The resource established under this project will enable a wide array of future and prospective migraine-related studies. For example, the migraine risk assessment and prediction developed here are candidates for prospective validation studies. Given that migraine has a strong genetic component, the migraine library can easily incorporate patients' genomic and epigenetic information as it becomes available.

Migraine is a woefully underserved condition. There is one neurologist for 1 million Pakistanis[16]. MigraineAid will help patients do better self-care, and the chat interface will allow non-migraine specialists to provide specialist level care thereby acting as a <u>*useful healthcare extender*</u>. Our relationships with Indus Hospital and Health Network (IHHN), one of the largest non-governmental health networks in Pakistan gives us the reach to distribute the MPM widely once we have proven that it can help migraineurs.

The framework developed for migraine prediction and prevention will be applicable to other CDEE like depression, irritable bowel disease, multiple sclerosis, rheumatoid arthritis, and asthma to name a few. The human suffering and economic burden associated with CDEE are enormous, and a framework that not only empowers the patients to take charge of their own disease but also helps them manage it more effectively will give rise to an entirely new approach in management of CDEE.

Project timeline
To the right, we outline the approximate timeline of the project, noting that dynamic adjustments to the schedule might always be necessary to ensure that unanticipated obstacles be dealt with in an expeditious manner.



Bibliography and References

1. Ashina M, Katsarava Z, Do TP, Buse DC, Pozo-Rosich P, Özge A, et al. Migraine: epidemiology and systems of care. Lancet. 2021;397(10283):1485–95. Available from: http://dx.doi.org/10.1016/s0140-6736(20)32160-7

2. Dagenais GR, Gerstein HC, Zhang X, McQueen M, Lear S, Lopez-Jaramillo P, et al. Variations in diabetes prevalence in low-, middle-, and high-income countries: Results from the prospective urban and rural epidemiological study. Diabetes Care. 2016;39(5):780–7. Available from: http://dx.doi.org/10.2337/dc15-2338

3. Lipton RB, Bigal ME, Diamond M, Freitag F, Reed ML, Stewart WF. Migraine prevalence, disease burden, and the need for preventive therapy. Neurology. 2007 [cited 2024 Mar 11];68(5):343–9. Available from: https://pubmed.ncbi.nlm.nih.gov/17261680/

4. Amiri P, Kazeminasab S, Nejadghaderi SA, Mohammadinasab R, Pourfathi H, Araj-Khodaei M, et al. Migraine: A review on its history, global epidemiology, risk factors, and comorbidities. Front Neurol. 2021;12:800605. Available from: http://dx.doi.org/10.3389/fneur.2021.800605

5. Vetvik KG, MacGregor EA. Sex differences in the epidemiology, clinical features, and pathophysiology of migraine. Lancet Neurol. 2017;16(1):76–87. Available from: http://dx.doi.org/10.1016/S1474-4422(16)30293-9

6. Buse DC, Manack AN, Fanning KM, Serrano D, Reed ML, Turkel CC, et al. Chronic migraine prevalence, disability, and sociodemographic factors: results from the American Migraine Prevalence and Prevention Study. Headache. 2012;52(10):1456–70. Available from: http://dx.doi.org/10.1111/j.1526-4610.2012.02223.x

7. Morgan R, Ayiasi RM, Barman D, Buzuzi S, Ssemugabo C, Ezumah N, et al. Gendered health systems: evidence from low- and middle-income countries. Health Res Policy Syst. 2018;16(1). Available from: http://dx.doi.org/10.1186/s12961-018-0338-5

8. Goadsby PJ, Lipton RB, Ferrari MD. Migraine--current understanding and treatment. N Engl J Med. 2002;346(4):257–70. Available from: http://dx.doi.org/10.1056/NEJMra010917

9. Ferrari MD. The economic burden of migraine to society. Pharmacoeconomics. 1998;13(6):667–76. Available from: http://dx.doi.org/10.2165/00019053-199813060-00003

10. Stewart WF, Lipton RB, Simon D. Work-related disability: results from the American migraine study. Cephalalgia. 1996;16(4):231–8; discussion 215. Available from: http://dx.doi.org/10.1046/j.1468-2982.1996.1604231.x

11. Michel P, Dartigues JF, Lindousli A, Henry P. Loss of productivity and quality of life in migraineurs among French workers: results from the GAZEL cohort. Headache. 1997;37:71–8.

12. Edmeads J, Findlay H, Tugwell P, Pryse-Phillips W, Nelson RF, Murray TJ. Impact of migraine and tension-type headache on life-style, consulting behaviour, and medication use: a Canadian population survey. Can J Neurol Sci. 1993;20(2):131–7. Available from: http://dx.doi.org/10.1017/s0317167100047697

13. Winnem J. Prevalence of adult migraine in general practice. Cephalalgia. 1992;12(5):300–3. Available from: http://dx.doi.org/10.1046/j.1468-2982.1992.1205300.x

14. Cottrell CK, Drew JB, Waller SE, Holroyd KA, Brose JA, O'Donnell FJ. Perceptions and needs of patients with migraine: a focus group study. Headache. 2003;43(4):428–428. Available from: http://dx.doi.org/10.1046/j.1526-4610.2003.03085_17.x

15. Cameron C, Kelly S, Hsieh S-C, Murphy M, Chen L, Kotb A, et al. Triptans in the acute treatment of migraine: A systematic review and network meta-analysis. Headache. 2015;55 Suppl 4(S4):221–35. Available from: http://dx.doi.org/10.1111/head.12601

16. Farooq A, Venketasubramanian N, Wasay M. Stroke care in Pakistan. Cerebrovasc Dis Extra. 2021;11(3):118–21. Available from: http://dx.doi.org/10.1159/000519554

17. Gago-Veiga A, Pagán J, Henares K, Heredia P, González-García N, De Orbe M-I, et al. To what extent are patients with migraine able to predict attacks? J Pain Res. 2018;11:2083–94. Available from: http://dx.doi.org/10.2147/jpr.s175602

18. Casanova A, Vives-Mestres M, Donoghue S, Mian A, Martin PR. An observational study of self-reported migraine triggers and prospective evaluation of the relationships with occurrence of attacks enabled by a smartphone application (App). Headache. 2022;62(10):1406–15. Available from: http://dx.doi.org/10.1111/head.14328

19. Hoffmann J, Recober A. Migraine and triggers: post hoc ergo propter hoc? Curr Pain Headache Rep. 2013;17(10):370. Available from: http://dx.doi.org/10.1007/s11916-013-0370-7

20. Houle TT, Turner DP. Natural experimentation is a challenging method for identifying headache triggers. Headache. 2013;53(4):636–43. Available from: http://dx.doi.org/10.1111/head.12075

21. Lipton RB, Pavlovic JM, Haut SR, Grosberg BM, Buse DC. Methodological issues in studying trigger factors and premonitory features of migraine. Headache. 2014;54(10):1661–9. Available from: http://dx.doi.org/10.1111/head.12464

22. Pavlovic JM, Buse DC, Sollars CM, Haut S, Lipton RB. Trigger factors and premonitory features of migraine attacks: summary of studies. Headache. 2014;54(10):1670–9. Available from: http://dx.doi.org/10.1111/head.12468

23. Peroutka SJ. What turns on a migraine? A systematic review of migraine precipitating factors. Curr Pain Headache Rep. 2014;18(10):454. Available from: http://dx.doi.org/10.1007/s11916-014-0454-z

24. Seng EK. Headache triggers and causality editorial: a research roadmap and clinical caution. Headache. 2013;53(4):584–6. Available from: http://dx.doi.org/10.1111/head.12073

25. Turner DP, Smitherman TA, Martin VT, Penzien DB, Houle TT. Causality and headache triggers. Headache. 2013;53(4):628–35. Available from: http://dx.doi.org/10.1111/head.12076

26. https://www.ninds.nih.gov/sites/default/files/2023-03/NINDS_Strategic_Plan_2021-2026_Final_508C.pdf. Accessed March 9th, 2024

27. Houle TT, Turner DP, Golding AN, Porter JAH, Martin VT, Penzien DB, et al. Forecasting individual headache attacks using perceived stress: Development of a multivariable prediction model for persons with episodic migraine. Headache. 2017;57(7):1041–50. Available from: http://dx.doi.org/10.1111/head.13137

28. Stubberud A, Ingvaldsen SH, Brenner E, Winnberg I, Olsen A, Gravdahl GB, et al. Forecasting migraine with machine learning based on mobile phone diary and wearable data. Cephalalgia. 2023;43(5):3331024231169244. Available from: http://dx.doi.org/10.1177/03331024231169244

29. Sano A, Picard RW. Stress Recognition using Wearable Sensors and Mobile Phones. 2013 Humaine Association Conference on Affective Computing and Intelligent Interaction (Acii). 2013:671-676.

30. Silva-Néto RP, de Almeida Soares A, Augusto Carvalho de Vasconcelos C, da Silva Lopes L. Watermelon and others plant foods that trigger headache in migraine patients. Postgrad Med. 2021;133(7):760–4. Available from: http://dx.doi.org/10.1080/00325481.2021.1922211

31. Gazerani P. Migraine and diet. Nutrients. 2020;12(6):1658. Available from: http://dx.doi.org/10.3390/nu12061658

32. Hindiyeh NA, Zhang N, Farrar M, Banerjee P, Lombard L, Aurora SK. The role of diet and nutrition in migraine triggers and treatment: A systematic literature review. Headache. 2020;60(7):1300–16. Available from: http://dx.doi.org/10.1111/head.13836

33. Sun-Edelstein C, Mauskop A. Foods and supplements in the management of migraine headaches. Clin J Pain. 2009;25(5):446–52. Available from: http://dx.doi.org/10.1097/ajp.0b013e31819a6f65

34. Spekker E, Nagy-Grócz G. All roads lead to the gut: The importance of the Microbiota and diet in migraine. Neurol Int. 2023;15(3):1174–90. Available from: http://dx.doi.org/10.3390/neurolint15030073

35. Hoffmann J, Schirra T, Lo H, Neeb L, Reuter U, Martus P. The influence of weather on migraine - are migraine attacks predictable? Ann Clin Transl Neurol. 2015;2(1):22–8. Available from: http://dx.doi.org/10.1002/acn3.139

36. Yang AC, Fuh J-L, Huang NE, Shia B-C, Peng C-K, Wang S-J. Temporal associations between weather and headache: analysis by empirical mode decomposition. PLoS One. 2011;6(1):e14612. Available from: http://dx.doi.org/10.1371/journal.pone.0014612

37. Yang AC, Fuh J-L, Huang NE, Shia B-C, Wang S-J. Patients with migraine are right about their perception of temperature as a trigger: time series analysis of headache diary data. J Headache Pain. 2015;16(1). Available from: http://dx.doi.org/10.1186/s10194-015-0533-5

38. Empatica. https://www.empatica.com/. Accessed March 9, 2024.

39. Herekar AA, Ahmad A, Uqaili UL, Ahmed B, Effendi J, Alvi SZ, et al. Primary headache disorders in the adult general population of Pakistan – a cross sectional nationwide prevalence survey. J Headache Pain. 2017;18(1). Available from: http://dx.doi.org/10.1186/s10194-017-0734-1

40. https://www.pta.gov.pk/en/telecom-indicators. Accessed March 9th, 2024

41. https://gs.statcounter.com/os-market-share/mobile/pakistan. Accessed March 9th, 2024

42. https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=PK. Accessed March 12th, 2024

43. https://www.c-salt.org/downloads/prus. Accessed March 9th, 2024

44. Biovotion AG. Everion 2018 p.1-4 Google Scholar

45. Sano A, Taylor S, McHill AW, Phillips AJK, Barger LK, Klerman E, et al. Identifying objective physiological markers and modifiable behaviors for self-reported stress and mental health status using wearable sensors and mobile phones: Observational study. J Med Internet Res. 2018;20(6):e210. Available from: http://dx.doi.org/10.2196/jmir.9410

46. Spierings ELH, Donoghue S, Mian A, Wöber C. Sufficiency and Necessity in Migraine: How do we Figure Out if Triggers are Absolute or Partial and, if Partial, Additive or Potentiating? Curr Pain Headache Rep. 2014;18(10). Available from: http://dx.doi.org/10.1007/s11916-014-0455-y

47. Pucci E, Taino G, Imbriani M, Costa A, Cristina S, Antonaci F. P033. Headache and commuting: preliminary data in a group of workers. J Headache Pain. 2015;16(S1). Available from: http://dx.doi.org/10.1186/1129-2377-16-s1-a79

48. Mehndiratta M, Garg D, Wasay M, Aggarwal V. Air pollution and headache disorders. Ann Indian Acad Neurol. 2022;25(7):35. Available from: http://dx.doi.org/10.4103/aian.aian_1138_21

49. Blabst N, Diefenbach S. WhatsApp and Wellbeing: A study on WhatsApp usage, communication quality and stress. In: Electronic Workshops in Computing. BCS Learning & Development; 2017.

50. Li Y, Mamouei M, Salimi-Khorshidi G, Rao S, Hassaine A, Canoy D, et al. Kazem Rahimi Hi-BEHRT: Hierarchical Transformer-based model for accurate prediction of clinical events using multimodal longitudinal electronic health records.

51. Ferroni P, Barbanti P, Della-Morte D, Palmirotta R, Jirillo E, Guadagni F. Redox mechanisms in migraine: Novel therapeutics and dietary interventions. Antioxid Redox Signal. 2018;28(12):1144–83. Available from: http://dx.doi.org/10.1089/ars.2017.7260

52. Dhamyal H, Ali A, Ihsan Ayyub Qazi AA. Using Self Attention DNNs to Discover Phonemic Features for Audio Deep Fake Detection, ASRU 2021 - IEEE Automatic Speech Recognition and Understanding workshop. Cartagena, Colombia; 2021.

53. Ali A, Raza AA, Qazi IA. Validated digital literacy measures for populations with low levels of internet experiences. Dev Eng [Internet]. 2023;8(100107):100107. Available from: http://dx.doi.org/10.1016/j.deveng.2023.100107

54. IMS Institute for Healthcare Informatics. Patient Adoption of mHealth – Use, Evidence and Remainig Barriers to Mainstream Acceptance. Pasippany, NJ, USA: IMS Health Incorporated; 2015.

55. Ashina M, Terwindt GM, Al-Karagholi MA-M, de Boer I, Lee MJ, Hay DL, et al. Migraine: disease characterisation, biomarkers, and precision medicine. Lancet [Internet]. 2021;397(10283):1496–504. Available from: http://dx.doi.org/10.1016/s0140-6736(20)32162-0

56. Kesavan P, Satheesh AP, Husain RSRA, Veerappan U, Kannaian S, Ahmed SS, Veerabathiran R. Genetic predisposition of TNFα gene polymorphism in South-Indian Migraineurs and meta-analysis. Front Biosci (Elite Ed). 2021 Dec 20;13(2):226-236. Available from: http://dx.doi.org/10.52586/e880

57. Papasavva M, Vikelis M, Siokas V, Katsarou M-S, Dermitzakis EV, Raptis A, et al. Variability in oxidative stress-related genes (SOD2, CAT, GPX1, GSTP1, NOS3, NFE2L2, and UCP2) and susceptibility to migraine clinical phenotypes and features. Front Neurol [Internet]. 2023;13. Available from: http://dx.doi.org/10.3389/fneur.2022.1054333

58. Nappi G, Costa A, Tassorelli C, Santorelli FM. Migraine as a complex disease: heterogeneity, comorbidity and genotype-phenotype interactions. Funct Neurol. 2000;15(2):87–93.

59. Buse DC, Manack AN, Fanning KM, Serrano D, Reed ML, Turkel CC, et al. Chronic migraine prevalence, disability, and sociodemographic factors: results from the American Migraine Prevalence and Prevention Study. Headache. 2012;52(10):1456–70. Available from: http://dx.doi.org/10.1111/j.1526-4610.2012.02223.x

60. Le H, Tfelt-Hansen P, Russell MB, Skytthe A, Kyvik KO, Olesen J. Co-morbidity of migraine with somatic disease in a large population-based study. Cephalalgia. 2011;31(1):43–64. Available from: http://dx.doi.org/10.1177/0333102410373159